

Variability in Young Children’s Everyday Visual Experiences of Object Categories

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Abstract

Children learn to categorize objects from everyday visual experience. How variable are the exemplars and views of objects in that experience – for example, of birds, cribs, and cups? Here, we quantified the variability of the object categories detected in videos taken from the child’s perspective in the BabyView dataset (Long et al., 2024). Detected objects incorporate both viewpoint variability (the range of angles and scales at which an object is seen) and exemplar variability (the range of distinct instances within a category). Using embeddings from both a vision–language model (CLIP; Radford et al., 2021) and a self-supervised vision model (DINOv3; Simeoni et al., 2025), we quantified global dispersion (mean distance from each exemplar to its within-category centroid) and local dispersion (mean within-category k -nearest-neighbor distance, $k = 5$) on the same set of 7,018 human-validated crops spanning 85 CDI noun categories. Both metrics varied widely across categories, with moderate model agreement moderate for global dispersion ($\rho = .55$) and local dispersion ($\rho = .63$). Dispersion metrics were weakly associated with how frequently a category was detected in egocentric video: categories could appear equally often but differ in visual variability. Our results motivate future work that examines how category frequency and variability interact to yield robust category representations during everyday learning in early childhood.

Introduction

To learn a visual category – for example, a bird – children need to extract visual features from the exemplars they experience in everyday learning environments. Recent work has begun characterizing the statistics of this input, showing that object categories follow a skewed frequency distribution in the infant view, with a small number of categories dominating and most appearing rarely (Clerkin et al., 2017; Yang et al., 2026). Intriguingly, the words for frequently viewed categories tend to be learned earlier in development (Clerkin et al., 2017).

But any two categories can appear equally often while differing in how variable their exemplars are – both in the range of distinct instances (e.g., many different cups versus one recurring cup) and the range of viewpoints,

poses, and contexts in which those instances appear. Experimental work shows that both kinds of variability matter for human category learning: training with more variable exemplars slows initial learning but improves category generalization (Posner & Keele, 1968; Raviv et al., 2022), and toddlers trained with perceptually variable exemplars generalize category names more broadly and show accelerated vocabulary growth (Perry et al., 2010). Yet at present, we know little about the actual variability of the exemplars that learners encounter during early childhood. Here, we analyze the variability in object categories detected in egocentric videos taken from the child’s perspective (Long et al., 2024).

Methods

Dataset and model embeddings We analyzed a dataset of egocentric videos from early childhood (Long et al., 2024; Yang et al., 2026). We detected candidate objects using the YOLOE model (Wang et al., 2025) and then human-validated a subset of detections from 85 categories ($N=7,018$ images, 57–100 crops per category) that correspond to nouns from the CDI vocabulary checklist (Marchman & Dale, 2023). We extracted per-crop embeddings for these human-validated detections from CLIP ViT-B/32 and DINOv3 ViT-B/16.

Global dispersion (distance-to-centroid) For each category c , we took the mean L2 distance from each exemplar to its category centroid:

$$V_c = \frac{1}{|c|} \sum_{x_i \in c} d(x_i, \mu_c),$$

where μ_c is the mean embedding of all exemplars in category c . Larger V_c indicates greater global dispersion.

Local dispersion (kNN distance) For each exemplar, we computed mean L2 distance to its five nearest neighbors ($k = 5$) within the same category. Larger values indicate greater local dispersion.

Proportion of detections For each category, we computed the proportion of detections in BabyView (release 2025.1) (Yang et al., 2026) as the number of detections in that category divided by the total number of



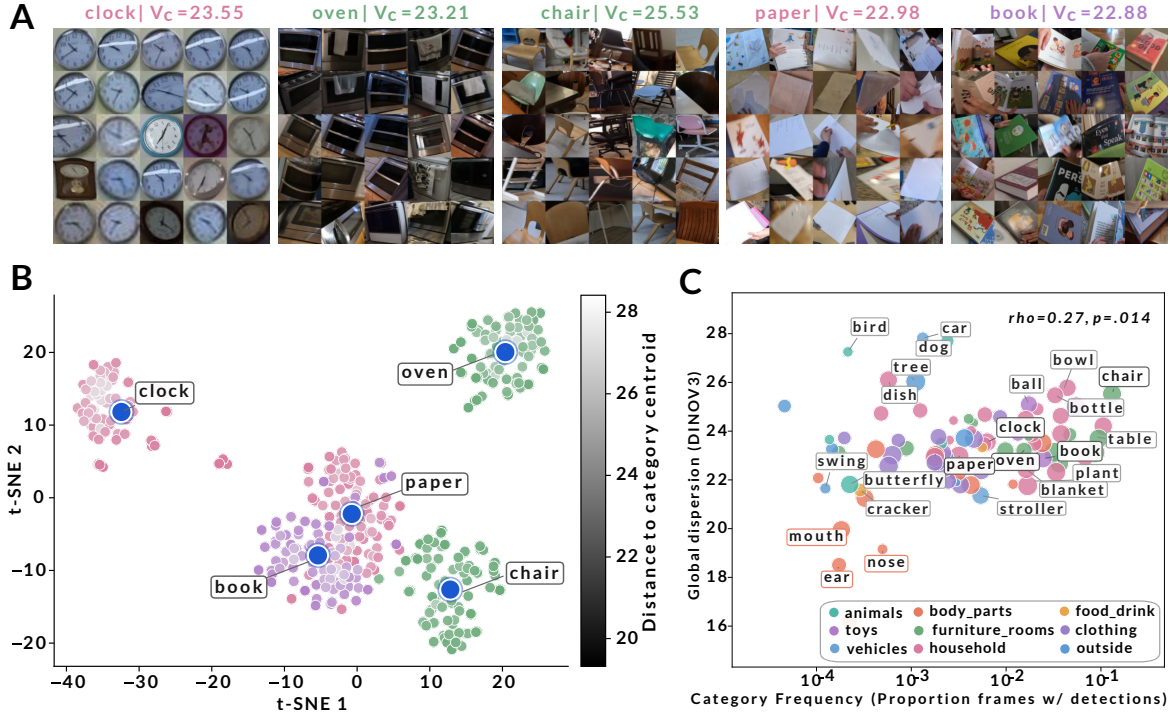


Figure 1: (A) Five selected categories ordered by increasing global dispersion (*clock*, *oven*, *chair*, *paper*, *book*). (B) DINOv3 t-SNE of a few selected categories, highlighting variability; dots are colored by their category and distance to category centroid. (C) $\log_{10}$ category frequency vs. global dispersion (DINOv3).

frames in the dataset. Figure 1C plots $\log_{10}$ proportion of detections; Spearman ρ uses untransformed values.

Results

We found that global dispersion values were widely distributed across all 85 categories (CLIP: mean = 18.12, SD = 1.84, range [13.10, 21.59]; DINOv3: mean = 23.36, SD = 1.76, range [16.18, 27.81]).¹ These resulted in visible differences in category heterogeneity (Figure 1A–B). Categories such as *car* ($V_c = 27.81$) and *bird* ($V_c = 27.25$) had among the highest dispersion, while *pants* ($V_c = 21.79$) and *pajamas* ($V_c = 22.71$) among the lowest dispersion.² We found the same pattern of effects using CLIP ViT, with moderate agreement between CLIP and DINOv3 (Spearman $\rho = .55, p < .001$). We also found convergence in local dispersion metrics: categories varied in local kNN dispersion (DINOv3: mean = 27.05, SD = 2.60, range [17.49, 33.29]), again with moderate agreement across CLIP and DINOv3 (Spearman $\rho = .63, p < .001$). Finally, we examined the degree to which categories that were highly variable appeared often in the infant view: we found that these metrics were

¹Note that absolute means are not directly comparable across models because embedding dimensionality and scale differ.

²While body-part detections have low dispersion, not included in montages to protect participant privacy.

only weakly associated, highlighting their distinct contributions to learning (correlation between detection frequency and global dispersion: Spearman $\rho = .18, p = .095$; DINOv3: Spearman $\rho = .26, p = .014$; Figure 1C).

Discussion

In naturalistic settings, infants encounter some categories in consistent configurations (e.g., a crib seen mostly from the same angles) and other categories in varying contexts, poses, and instances (e.g., toy cars seen from many viewpoints). We quantified this within-category variability in the object categories that young children experience. As exemplar variability modulates how quickly a category can be learned and generalized (Perry et al., 2010), these results make new predictions about which categories may be easier or harder for children to learn. However, our metrics do not distinguish between different sources of exemplar variability; developing new quantitative metrics and relating these to learning outcomes remains an exciting avenue for future work. There are sampling limitations to the present work: these data are from mostly Western, high SES households, limiting generalizability. Nevertheless, these results motivate the need to consider and quantify exemplar variability to inform both our theories and models of category learning.

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